

ESP32 LED Controller — Complete Video Presentation Script

Scene-by-Scene Script, Voiceover Narration & Technical Demonstration Manual

1. Executive Overview

This script accompanies the interactive animated presentation file `video_guide.html`. It serves as a complete video presenter guide for demonstrating the ESP32 LED Matrix Web Controller, firmware architecture in `src/main.cpp`, parallel FastLED driving, and web interface controls.

Scene 1: Introduction & Architecture (0:00 - 0:25)

■ ■ Presenter Voiceover Narration:

"Hello! In this presentation, we explore the ESP32 LED Matrix Controller project. This firmware enables high-speed streaming and playback of RGB animation binary files (.bin) across WS2812B matrix displays using parallel data channels and a built-in web portal."

Scene 2: Firmware Code Structure (main.cpp) (0:25 - 1:00)

■ ■ Presenter Voiceover Narration:

"Let me walk you through the firmware code in `main.cpp`. The C++ project is built using PlatformIO and FastLED. We configure 4 parallel output controllers mapped to safe GPIO pins D2, D4, D5, and D12. LittleFS mounts the SPIFFS flash partition to store animation files."

Scene 3: Connecting to Web App (1:00 - 1:20)

■ ■ Presenter Voiceover Narration:

"When powered on, the ESP32 creates its own Wi-Fi Access Point network named 'ESP32-LED-Controller' (Password: 12345678). Once connected, open `http://192.168.4.1` in any browser to access the dynamic control interface."

Scene 4: Configuration & Sliders (1:20 - 1:50)

■ ■ Presenter Voiceover Narration:

"On the web app, you can adjust the Number of LED Strips and LEDs per Strip using live sliders or text boxes. The web app instantly calculates total pixel count, max amperage demand, and FastLED hardware refresh rates."

Scene 5: Smart Wiring Guide & Pin Allocations (1:50 - 2:15)

■ ■ Presenter Voiceover Narration:

"The Smart Wiring section features an interactive ESP32 board graphic with glowing pin highlights. It generates a live Pin Allocation Table showing which GPIO pin drives each strip section, plus common ground instructions."

Scene 6: Binary File Upload & Playback (2:15 - 2:30)

■ ■ Presenter Voiceover Narration:

"Finally, drag and drop any .bin animation file into the upload zone. The web server streams the file in 256KB chunks to LittleFS flash and immediately loops playback at 30 FPS!"